

Three Views of a Limit

Answer Key

Precalculus / Introduction to Limits

Quick Answers

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|----------------------------------|-------------------------|
| 1. 6 | 7. 2, 3 |
| 2. 8 | 8. 6 |
| 3. 4 | 9. $\frac{1}{6}$ |
| 4. 25 | 10. 2.0505, 2.005, 2 |
| 5. 0.4988, 0.5013, $\frac{1}{2}$ | 11. 3 |
| 6. 3 | 12. $\frac{31}{32}$, 1 |
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1. Numerical — read the trend, both sides.

From the left: 5.9, 5.99, 5.999. **From the right:** 6.1, 6.01, 6.001. Both columns close in on the same number, so the limit is that number. 6

Expected sentence: the limit describes where f is heading as x approaches 3, and the definition deliberately never evaluates f at 3 itself. Since $f(x) = x + 3$ for every $x \neq 3$, the approach is perfectly ordinary even though $f(3)$ is undefined — a hole in the graph, not a break in the trend.

2. Algebraic — factor and cancel.

Step 1. Substitution gives $\frac{0}{0}$, an indeterminate form, so rewrite rather than conclude.

Step 2. Difference of squares: $\frac{x^2 - 16}{x - 4} = \frac{(x - 4)(x + 4)}{x - 4} = x + 4$ for $x \neq 4$.

Step 3. Now substitution is legal: $4 + 4$. 8

3. Graphical — a hole does not stop a limit.

Read the graph. As x moves toward 2 from the left the curve rises toward height 4; from the right it falls toward the same height 4. Both sides agree. 4

Expected sentence: the open circle records that $g(2)$ is undefined, which is a statement about the single point $x = 2$. The limit is about the values of g at inputs *near* 2, and those are unaffected. Algebraically $g(x) = x + 2$ for $x \neq 2$, which is why the graph is a straight line with one point removed.

4. Series — a finite partial sum.

Expand the sigma notation by substituting $n = 1, 2, 3, 4, 5$:

$$\sum_{n=1}^5 (2n - 1) = 1 + 3 + 5 + 7 + 9$$

Add: $1 + 3 = 4$, $+5 = 9$, $+7 = 16$, $+9 = 25$. 25

Worth noticing: the running totals are 1, 4, 9, 16, 25 — the perfect squares. The k th partial sum of the odd numbers is k^2 .

5. Numerical — build the table, then name the exact value.

Left entry ($x = -0.01$): $\frac{\sqrt{0.99} - 1}{-0.01} = \frac{-0.0050126\dots}{-0.01} = 0.5012562\dots$ 0.5013

Right entry ($x = 0.01$): $\frac{\sqrt{1.01} - 1}{0.01} = \frac{0.0049876\dots}{0.01} = 0.4987562\dots$ 0.4988

The exact limit. The two entries squeeze in on 0.5 from opposite sides. Confirm it algebraically with the conjugate: $\frac{\sqrt{x+1} - 1}{x} \cdot \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} = \frac{x}{x(\sqrt{x+1} + 1)} = \frac{1}{\sqrt{x+1} + 1}$, which at $x = 0$ is $\frac{1}{1+1}$. $\frac{1}{2}$

Grading note: the table alone is *evidence*, not proof — it can never rule out something happening between the sampled points. The algebra is what settles it.

6. Algebraic — cubes over squares.

Step 1. Substitution gives $\frac{0}{0}$; both top and bottom vanish at $x = 2$, so both carry a factor of $(x - 2)$.

Step 2. Factor each:

$$\frac{x^3 - 8}{x^2 - 4} = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)} = \frac{x^2 + 2x + 4}{x + 2} \quad (x \neq 2)$$

Step 3. Substitute $x = 2$: $\frac{4 + 4 + 4}{4} = \frac{12}{4}$. 3

7. Graphical — one-sided limits and a jump.

(a) **From the left** the graph follows the line through $(0, 1)$ rising to the open circle at height 2: 2

(b) **From the right** the graph starts at the solid dot at height 3 and falls away: 3

(c) **Manual review — expected justification.** The two-sided limit exists only when both one-sided limits exist *and* are equal. Here they are 2 and 3, which are not equal, so $\lim_{x \rightarrow 1} h(x)$ does **not** exist. Accept any answer that compares the two one-sided values; do not accept “because the graph breaks” with no reference to them. Note also that $h(1) = 3$ exists — a function can have a value at a point where the limit does not, which is the mirror image of problem 3.

8. Series — an infinite geometric sum.

Identify the pieces. First term $a = 3$ (take $n = 0$), common ratio $r = \frac{1}{2}$, and $|r| < 1$, so the series converges.

Apply the formula $S = \frac{a}{1 - r}$: $S = \frac{3}{1 - \frac{1}{2}} = \frac{3}{\frac{1}{2}}$. 6

The partial-sum limit it represents. $S_k = \sum_{n=0}^k 3 \left(\frac{1}{2}\right)^n = 6 \left(1 - \left(\frac{1}{2}\right)^{k+1}\right)$, and the value above is $\lim_{k \rightarrow \infty} S_k$. The sum of an infinite series is defined as that limit.

9. Algebraic — rationalize the numerator.

Step 1. Substitution gives $\frac{0}{0}$, and the trouble is a radical, so multiply by the conjugate of the numerator.

Step 2.

$$\frac{\sqrt{x+9}-3}{x} \cdot \frac{\sqrt{x+9}+3}{\sqrt{x+9}+3} = \frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \frac{1}{\sqrt{x+9}+3} \quad (x \neq 0)$$

Step 3. Substitute $x = 0$: $\frac{1}{3+3}$.

$\frac{1}{6}$

10. Numerical — end behaviour from a table.

At $x = 100$: $\frac{2(100)+3}{100-1} = \frac{203}{99} = 2.0505050\dots$

2.0505

At $x = 1000$: $\frac{2(1000)+3}{1000-1} = \frac{2003}{999} = 2.0050050\dots$

2.0050

The limit. The values are marching down toward 2. Algebraically, divide top and bottom by x : $\frac{2+3/x}{1-1/x}$, and both $3/x$ and $1/x$ go to 0 as x grows.

2

11. Graphical then algebraic — a horizontal asymptote.

Read the graph. Far out in either direction the curve flattens against the dashed line, so the limit is the height of that line.

3

Confirm algebraically. Divide numerator and denominator by x^2 :

$$\frac{3x^2+1}{x^2+2} = \frac{3+\frac{1}{x^2}}{1+\frac{2}{x^2}} \longrightarrow \frac{3+0}{1+0}$$

Teaching note: a horizontal asymptote is a limit statement about end behaviour. The curve may cross an asymptote elsewhere; what matters is where it settles.

12. Synthesis — the sum of a series is a limit.

(a) **The fifth partial sum.** $S_5 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$. Over the common denominator 32 this is $\frac{16+8+4+2+1}{32}$.

$\frac{31}{32}$

(b) **The infinite sum.** Geometric with $a = \frac{1}{2}$ and $r = \frac{1}{2}$, so $S = \frac{1/2}{1-1/2}$.

1

(c) **Manual review — expected explanation.** The running totals are $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \frac{31}{32}$: each one closes half of the remaining gap to 1, so $S_k = 1 - (\frac{1}{2})^k$ and no partial sum ever reaches 1. Numerically the totals approach 1; graphically the plotted partial sums flatten against the line $y = 1$; algebraically $\lim_{k \rightarrow \infty} S_k = 1$. The sum of an infinite series is *defined* to be that limit — not the result of an endless addition, which is why an infinite sum can be a finite number at all. Accept any answer that names the sum as a limit of partial sums; do not accept “you keep adding forever”.